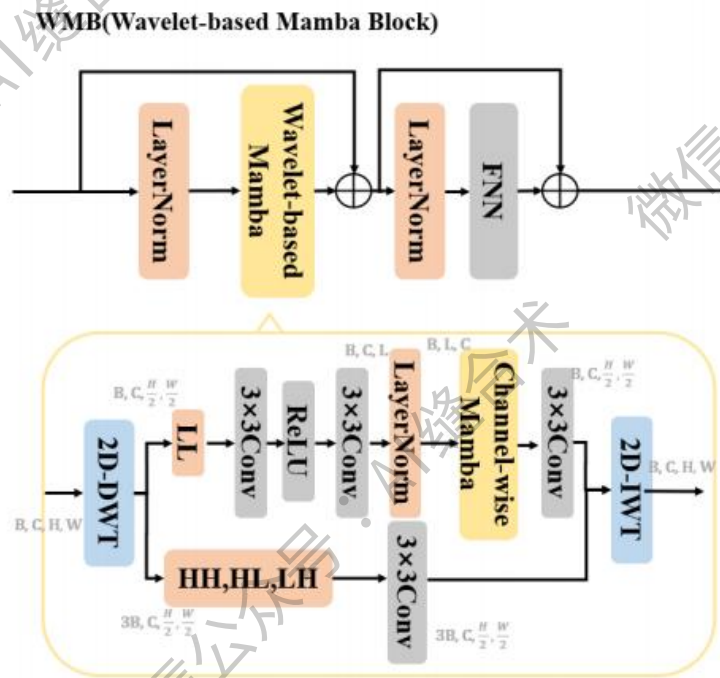


# 论文信息



**论文题目:** Wavelet-based Mamba with Fourier Adjustment for Low-light Image Enhancement

**中文题目:** 基于小波的 Mamba 与傅里叶调整用于低光图像增强

**论文链接:** <https://arxiv.org/pdf/2410.20314>

**官方 github:** <https://github.com/mcpaulgeorge/WalMaFa/tree/master>

**所属机构:** 上海理工大学、辽宁师范大学、吉林大学和国防科技大学

**核心速览:** 本文介绍了一种名为 WalMaFa 的新型低光图像增强模型，该模型结合了小波变换和快

速傅里叶变换（FFT）的优势，通过引入基于通道的 Mamba 模块和快速傅里叶调整块（FFAB），实现了对低光图像的全局亮度增强和局部纹理细节的精细调整。WalMaFa 模型采用编码器-潜在层-解码器结构，通过实验验证了其在性能、用户感知效果和计算复杂度方面的优越性。

```
)
(ffn): FeedForward(
  (rep_conv1): Conv2d_BN(
    (c): Conv2d(85, 85, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=85, bias=False)
    (bn): BatchNorm2d(85, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
  (rep_conv2): Conv2d_BN(
    (c): Conv2d(85, 85, kernel_size=(1, 1), stride=(1, 1), groups=85, bias=False)
    (bn): BatchNorm2d(85, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
  (project_in): Conv2d(32, 85, kernel_size=(1, 1), stride=(1, 1))
  (dwconv): Conv2d(85, 85, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=85)
  (project_out): Conv2d(85, 32, kernel_size=(1, 1), stride=(1, 1))
)
(mb): WM(
  (convb): Sequential(
    (0): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU()
    (2): Conv2d(64, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  )
  (model1): Mamba(
    (in_proj): Linear(in_features=32, out_features=128, bias=False)
    (conv1d): Conv1d(64, 64, kernel_size=(4,), stride=(1,), padding=(3,), groups=64)
    (act): SiLU()
    (x_proj): Linear(in_features=64, out_features=66, bias=False)
    (dt_proj): Linear(in_features=2, out_features=64, bias=True)
    (out_proj): Linear(in_features=64, out_features=32, bias=False)
  )
  (smooth): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (ln): LayerNorm((32,), eps=1e-05, elementwise_affine=True)
  (softmax): Softmax(dim=None)
)
)
)
微信公众号: AI缝合术!
Input shape: torch.Size([1, 32, 256, 256])
Output shape: torch.Size([1, 32, 256, 256])
```